👩‍🏫 Staff Guide · Chemistry · Unit 1C The Strike & Staying Safe (Level 2)

Level 2 = about 3rd-grade reading. SLE students stay in Mr. Brittle's lesson and use these pieces next to it: same topics, same order, same points.

1. How the pieces line up

Gen ed piece (Canvas: "Unit 1 Lightning: L1.10 Energy Flow, L1.11 The Strike, L1.12 Conductors & Insulators, L1.13 No Swimming, L1.13.5 You Light Up My Life, L1.14 Unit Test + Lightning Model")	**Level 2 piece**
L1.10 Energy Flow (Concord sim) + slides	Slides "Storm cloud" / "Energy flow" + notes part A
L1.11 The Strike (NOAA / NWS sources)	Slides "The strike" + notes part A
L1.12 Conductors & Insulators (references reading)	Slides "Lets it through or stops it" + notes part B
L1.13 No Swimming (salt water lab) + L1.13.5 You Light Up My Life (conductivity lab)	Conductivity lab packet + notes part C
L1.14 Practice Test, Unit Test (paper) + Lightning Model rubric	Unit quiz below; Lightning Model: Level 1-2 label the picture model in the notes, 🚧 check with Mr. Brittle
Class slides	Deck: 17 slides, Words to Know first, hold-up-1-2-3 checks, recap
His worksheets (grouped into lessons)	Guided notes: 14 items + key
Class quiz / notes check	Quiz: 10 questions + key
Vocabulary	8 picture cards (cut apart)
—	1 game on the class site: 10 questions + a matching round

2. Supports built in

- Short lines, key words bold, word bank in every notes box
- Quiz: 3 choices, no "all of the above"
- Picture anchors: 🔌 lets it through, 🧤 stops it, 🏠 inside = safe
- Safety rule to memorize: "When thunder roars, go indoors."

3. Add these as the IEP allows

- Read-aloud · extended time · separate setting
- Printed slides · videos paused at the "look for" part

4. What to grade

- Lightning Unit Test (paper, not in the export): our unit quiz stands in until Mr. Brittle shares the test
- **Notes**: same points as his worksheets for that lesson when complete.
- **Quiz**: score, then scale to the class points.
- These are **modifications** (less content). The IEP must allow modified grading. Check with the case manager.

5. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm in her files say "check with Mr. Brittle".

🚧 Under construction (check with Mr. Brittle)

- **L1.14 Lightning Unit Test (paper)**: Not in the export (only the practice test is). Our quiz is built from the practice test.
- **L1.14 Lightning Model (rubric)**: We did not build a modified model rubric; ask Mr. Brittle what Level 1-3 students should turn in.

6. Photo credits (open licenses)

Lightning 14.07.2009 20-42-33.JPG — Hansueli Krapf, CC BY-SA 3.0, Wikimedia Commons · Lightning rod, Woodland Plantation House, LaPlace, Louisiana.jpg — Infrogmation of New Orleans, CC BY-SA 4.0, Wikimedia Commons · Electric guide 3×2.5 mm.jpg — Petar Milošević, CC BY-SA 4.0, Wikimedia Commons

👩🏾‍🏫 Staff Guide · Conductivity Lab: What Lets Electricity Through? (Level 2)

Chemistry · Unit 1C · goes with the class “L1.13 No Swimming, L1.13.5 You Light Up My Life” (same points as the class lab). Level 2 = about 3rd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Lab card**: L1.13 No Swimming (salt water + light bulb) and L1.13.5 You Light Up My Life (conductivity tester with solids and solutions). 8 steps, one per box. **check with Mr. Brittle**: Which chemicals Mr. Brittle uses (the class list includes copper(II) sulfate and calcium chloride)
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 2 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- “Is the light on or off?”
- “Bright or dim?”
- “Does it let electricity through?”

3. Modifications

- Data table given
- No amps-to-charges math (Level 3 does it)
- Word bank for the claim

4. What to grade (same points as the class lab)

- Same points: data table + claim sentence
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver (SLE).

6. Photo credits (open licenses)

Electric guide 3×2.5 mm.jpg — Petar Milošević, CC BY-SA 4.0, Wikimedia Commons

⛈️ Lightning lessons 10-14 notes

Chemistry · Unit 1C The Strike & Staying Safe · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Energy and the strike (L1.10, L1.11)

1. Separated charge stores _____ energy.
2. In a strike it turns into heat, light and _____.
3. Number the strike (1-4):

_____ A stepped leader moves down

_____ The return stroke flashes

_____ Charge builds at the cloud bottom

_____ A streamer reaches up

Word bank: **potential** **sound**

B. Conductors and insulators (L1.12, L1.13.5)

4. A _____ lets electrons move easily.
5. An _____ does not let electrons move easily.
6. Circle the conductors:

copper · rubber · salt water · plastic

Word bank: **conductor** **insulator**

C. Salt water and safety (L1.13, L1.14)

7. More salt in water makes more _____.
8. A lightning rod sends charge to the _____.
9. Match the place to safe or not safe:

inside a building	not safe: tall object
swimming pool	not safe: water conducts
under a tall tree	safe

Word bank: **ground** **ions**

Level 2 version of the gen ed “L1.10-L1.14 worksheets + practice test.”

📖 The Strike & Staying Safe: Vocabulary Picture Cards (Level 2)

Name: _____

Cut on the dashed lines. Use the cards to match, sort, or quiz a partner.



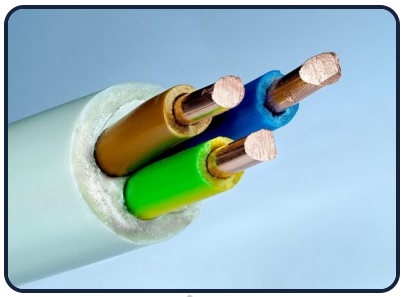
Energy

The ability to cause change. Stored (potential) energy in the cloud turns into heat, light and sound.



Thunder

The boom made when lightning heats the air so fast that it expands.



Conductor

A material that lets electrons move through it easily, like metal or salt water.



Insulator

A material that does not let electrons move easily, like rubber, plastic or dry wood.



Salt water

Salt breaks into charged particles (ions) in water, so salt water conducts electricity.



Lightning rod

A metal rod on a roof. It gives lightning a safe path to the ground.



Ground

The earth. Lightning's charge flows into the ground.

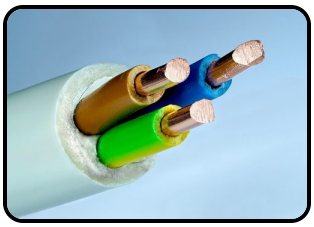


Current

The flow of charge (electrons or ions) through a material.

💡 Conductivity Lab: What Lets Electricity Through?

Chemistry · Unit 1C · Level 2



Name: _____

L1.13 No Swimming (salt water + light bulb) and L1.13.5 You Light Up My Life (conductivity tester with solids and solutions) · check with Mr. Brittle.

🧺 Test these

- ☐ 💧 **1 cup** plain water
- ☐ 🧂 **1 cup** salt water
- ☐ 🍬 **1 cup** sugar water
- ☐ 🔩 **1** piece of foil or copper
- ☐ 🕯️ **1** piece of wax (paraffin)

🌱 Steps: check each box when you finish

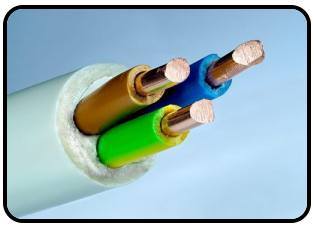
1.  Goggles on. Check the tester lights when its two ends touch. ☐
2.  **Predict**: will each thing conduct? Write yes or no. ☐
3.  **Test the solids**: foil/copper, wax, sugar, salt crystals. ☐
4.  **Test the liquids**: plain water, then salt water (1%, 5%, 10% if available). ☐
5.  **Test** sugar water. ☐
6.  **Record**: bright, dim, or off. ☐
7.  Dry the tester between every test. ☐
8.  **Sort**: conductors vs. insulators. What do the conductors have in common? ☐

📋 Data table

Material	Prediction	Bulb: bright / dim / off	Conductor?
foil / copper			
wax			
plain water			
1% salt water			
10% salt water			
sugar water			

💡 Conductivity Lab: What Lets Electricity Through? · Prep Sheet

Chemistry · Unit 1C · Level 2



Name: _____

👥 Job roles: circle your job. Initial when it is done.

Job	What you do	Done ✔️
💡 **Tester**	Touches the two tester ends to the material (both ends in the cup for liquids).	
👀 **Bulb watcher**	Says: bright, dim, or off.	
📝 **Recorder**	Writes the result in the table.	
🧻 **Dryer**	Dries the tester ends with a paper towel between tests.	

🧰 Tools and supplies: check when you have it

- ☐ 💡 **Conductivity tester** staff sets it up; low voltage
- ☐ 🥽 **Goggles** everyone
- ☐ 🧻 **Paper towels** dry the tester between tests
- ☐ 🧪 **Small cups** one per thing

⚠️ Safety and manners focus

🥽 Goggles on. Copper sulfate and salt solutions can sting eyes.

🚫 Never taste anything in a lab, even salt or sugar water.

🔌 Only use the low-voltage tester. Never put anything in a wall outlet.

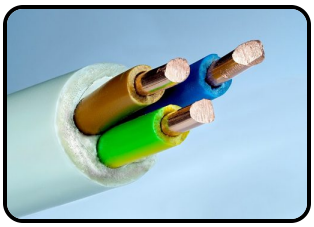
🧼 Wash hands after.

✅ Before you start

☐ Goggles on ☐ Tester is dry ☐ Table ready

💡 Conductivity Lab: What Lets Electricity Through? · Reflection

Chemistry · Unit 1C · Level 2



Name: _____

📸 My photos / proof

☐ Tester on metal ☐ Salt water bulb ☐ Sugar water bulb

✅ Skills I used today (check all you did)

☐ 💡 I tested each material

☐ 👀 I recorded bright, dim, or off

☐ 🔌 I sorted conductors and insulators

☐ 🥽 I wore goggles the whole lab

⭐ Which things were conductors? What do they have in common?

1. _____
2. _____

Ideas: **metals** **salt water** **ions**

🚤 Why should you get out of a pool in a storm?

1. _____
2. _____

Ideas: **water has ions** **conducts** **lightning**

✅ The Strike & Staying Safe Quiz

Chemistry · Unit 1C · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. 🔋 Charge separated in a cloud stores…

A. Ice | B. Potential energy | C. Light

2. 🌩️ Thunder is caused by…

A. Clouds hitting each other | B. Rain falling | C. Air heated so fast it expands

3. ⚡ The bright flash we see is the…

A. Return stroke | B. Stepped leader | C. Streamer

4. 🔌 A conductor…

A. Stops electrons | B. Has no atoms | C. Lets electrons move easily

5. 🧤 Which is an insulator?

A. Plastic | B. Copper | C. Salt water

6. 🔩 Metals are good conductors because their electrons…

A. Are stuck tight | B. Can move freely | C. Are missing

7. 🧂 Salt water conducts because salt makes…

A. Bubbles | B. Ions (charged particles) | C. Neutrons

8. 📈 More salt in the water makes the current…

A. Bigger | B. Smaller | C. Stop

9. 🏠 A lightning rod protects a house by…

A. Blocking rain | B. Making the house neutral forever | C. Giving charge a path to the ground

10. 🌳 The safest place in a storm is…

A. Inside a building | B. Under a tall tree | C. In a lake

⛈️ Lightning lessons 10-14 notes · ANSWER KEY

Chemistry · Unit 1C The Strike & Staying Safe · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Energy and the strike (L1.10, L1.11)

1. Separated charge stores **potential** energy.
2. In a strike it turns into heat, light and **sound**.
3. Number the strike (1-4):

2 A stepped leader moves down

4 The return stroke flashes

1 Charge builds at the cloud bottom

3 A streamer reaches up

Word bank: **potential** **sound**

B. Conductors and insulators (L1.12, L1.13.5)

4. A **conductor** lets electrons move easily.
5. An **insulator** does not let electrons move easily.
6. Circle the conductors:

copper rubber **salt water** plastic

Word bank: **conductor** **insulator**

C. Salt water and safety (L1.13, L1.14)

7. More salt in water makes more **ions**.
8. A lightning rod sends charge to the **ground**.
9. Match the place to safe or not safe:

(Key: inside a building → safe · swimming pool → not safe: water conducts · under a tall tree → not safe: tall object)

inside a building	not safe: tall object
swimming pool	not safe: water conducts
under a tall tree	safe

Word bank: **ground** **ions**

Level 2 version of the gen ed "L1.10-L1.14 worksheets + practice test."
🔑 Key written by our team from Mr. Brittle's materials, please check.

✅ The Strike & Staying Safe Quiz · ANSWER KEY

Chemistry · Unit 1C · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. 🔋 Charge separated in a cloud stores… L1.10

A. Ice | **B. Potential energy** | C. Light

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A. Inside a building | B. Under a tall tree | C. In a lake

Answers

1-B · 2-C · 3-A · 4-C · 5-A · 6-B · 7-B · 8-A · 9-C · 10-A

Reworded from the gen ed "L1.14 Lightning Unit Practice Test". Score: __ / 10.

🔑 *Key written by our team from Mr. Brittle's materials, please check.*